Existence of Every Term of OEIS A085245

A direct consequence of Dirichlet's theorem

Primary author (AI): ChatGPT (OpenAI)

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Abstract

For every n >= 1, infinitely many positive integers k make k*2^n+1 a squarefree semiprime. Thus the least such k defining A085245 always exists. This follows from a standard application of Dirichlet's theorem on primes in arithmetic progressions. The separate fixed-multiplier conjecture mentioned in the OEIS entry is not proved here.

1. A theorem on semiprimes

A squarefree semiprime is the product of two distinct primes. Theorem. Let m >= 2 and let a be an integer with gcd(a,m)=1. Infinitely many squarefree semiprimes N satisfy N = a (mod m).

Proof. Dirichlet's theorem guarantees infinitely many primes p = a (mod m) and infinitely many primes q = 1 (mod m). Fix one such prime q, and vary p through the infinitely many eligible primes distinct from q. Every N=pq is the product of two distinct primes, and N = a*1 = a (mod m). The products are distinct because p varies. QED.

2. Application to OEIS A085245

OEIS A085245 defines a(n) as the least positive integer k such that k*2^n+1 is a semiprime. Corollary. The integer a(n) exists for every n >= 1; indeed there are infinitely many eligible k.

Proof. Set m=2^n and a=1 in the preceding theorem. There exist infinitely many squarefree semiprimes N with N = 1 (mod 2^n). Each yields a positive integer k=(N-1)/2^n with k*2^n+1=N. Different N yield different k, so this set of positive k is infinite and therefore has a least element. QED.

3. Scope and AI disclosure

The OEIS entry separately conjectures that 78557*2^n+1 is semiprime for infinitely many n. Our result does not address that conjecture: k is allowed to vary with n. This well-definedness proof is a routine consequence of Dirichlet's theorem, not an original discovery.

Primary author (AI): ChatGPT (OpenAI). Mathematical drafting and preliminary proof checking were performed using AI; no independent peer review or guarantee of correctness is claimed.

References

[1] OEIS Foundation, A085245, https://oeis.org/A085245 (consulted 11 November 2026).

[2] Dirichlet's theorem on primes in arithmetic progressions (classical theorem).